

Family of Woman Ignored at
L.A. ER Files Suit

The Associated Press

In healthcare

time is of the essence

HEALTHCARE IN CRISIS
DOCTOR: LET'S CUT
HOSPITAL DELAYS

One in 5 on wait list

Dallas Hospital No Longer
Blaming Technical Glitch for
Sending Ebola Patient Home

ADD TIME ON GOOGLE

Shamaya Jackson claims her newborn was dis-
carded from the hospital despite having an illness that
eventually caused her to "suffer and die"

By [Cara Lynn Shultz](#)

Published on September 25, 2025 11:00AM EDT



Hospitals in England where
wait times are so long
that patients are being
sent home. Is yours one of them?

Lawsuit claims negligence
led to woman's death

By [Bridgette Fox](#), Staff Writer

Nov 19, 2025

HEALTH CARE

Three North Texas Hospitals
Penalized for High Readmission
Rate

A Remedy for Readmission

Solving Meridian Healthcare's Biggest Obstacle



Our Role + Understanding Stakeholders

Meridian Health

- For-profit hospital corporation
- Provide medical services while generating revenue for owners, shareholders
- Satisfy the contractual and financial obligations to its employees.

Customers

- Focus on Quality Outcomes: Prioritize the reduction of 30-day readmissions particularly for the same condition cases which is over 40% of urban hospital readmissions.
- Equitable Care Access: Utilize social determinants of health risk scores (housing, income) which impact recovery and long-term wellness.
- Proactive Follow Up: Increase engagement through discharge calls and post-hospitalization follow-ups, which statistically are shown to lower readmission rates.

Shareholders

- Look for cost-reduction and revenue-generation
- More inpatient & outpatient volumes = more revenue
- Increase number of private insurance patients
- Market expansion & physician recruitment

Employees

- Incentive Alignment: Ensure surgeons and clinicians are rewarded for long-term patient health instead of having inpatient volume.
- Analytical Support: Empower staff with a "Readmission Preventability Score" to help identify high-risk patients who require intensive medication and additional care.
- Operational Efficiency: Without compromising the quality of care or extending the length of stay optimize the supply chain cost per adjusted patient day.

Regulatory Bodies

- CMS/Joint Commission measures the frequency of readmissions in <30 days.
- Excess Readmission Ratio (ERR) > 1.0 leads to reduction Medicare payments up to 3%
- If the readmission is related and preventable, it counts towards the rate.

Problem Analysis

- A patient readmitted within 3 days does not generate additional profit.
- Hospital profit depends on each department's annual reimbursement rate, not admission volume alone.
- A readmission within 30 days increases the hospital's readmission rate.
- Higher readmission rates can reduce reimbursement and increase penalties.



Core Issues:

Hospital readmissions create a compounding financial and operational problem.

01.

- Directly increase operational costs while also triggering reimbursement penalties and cuts.

02.

- Reduce revenue opportunities as beds are occupied by returning patients rather than new admissions
- Limited bed availability can lead to “left without being seen” cases in the ER, resulting in additional lost revenue

03.

- Readmissions place strain on clinical resources, increase staff workload, and contribute to reputational damage through poorer performance metrics.

Dimensions of the Data

Clinical Factors

- Percent of 30 day readmission
- Number of beds over rural, urban, and suburban regions

Socioeconomic Data

- Count of unique insurance types per region
 - Mapped social determinant score over rural, urban, and suburban regions
- + studied in conjunction with socioeconomic data from the Black Wealth Data Center



Unique Dimension

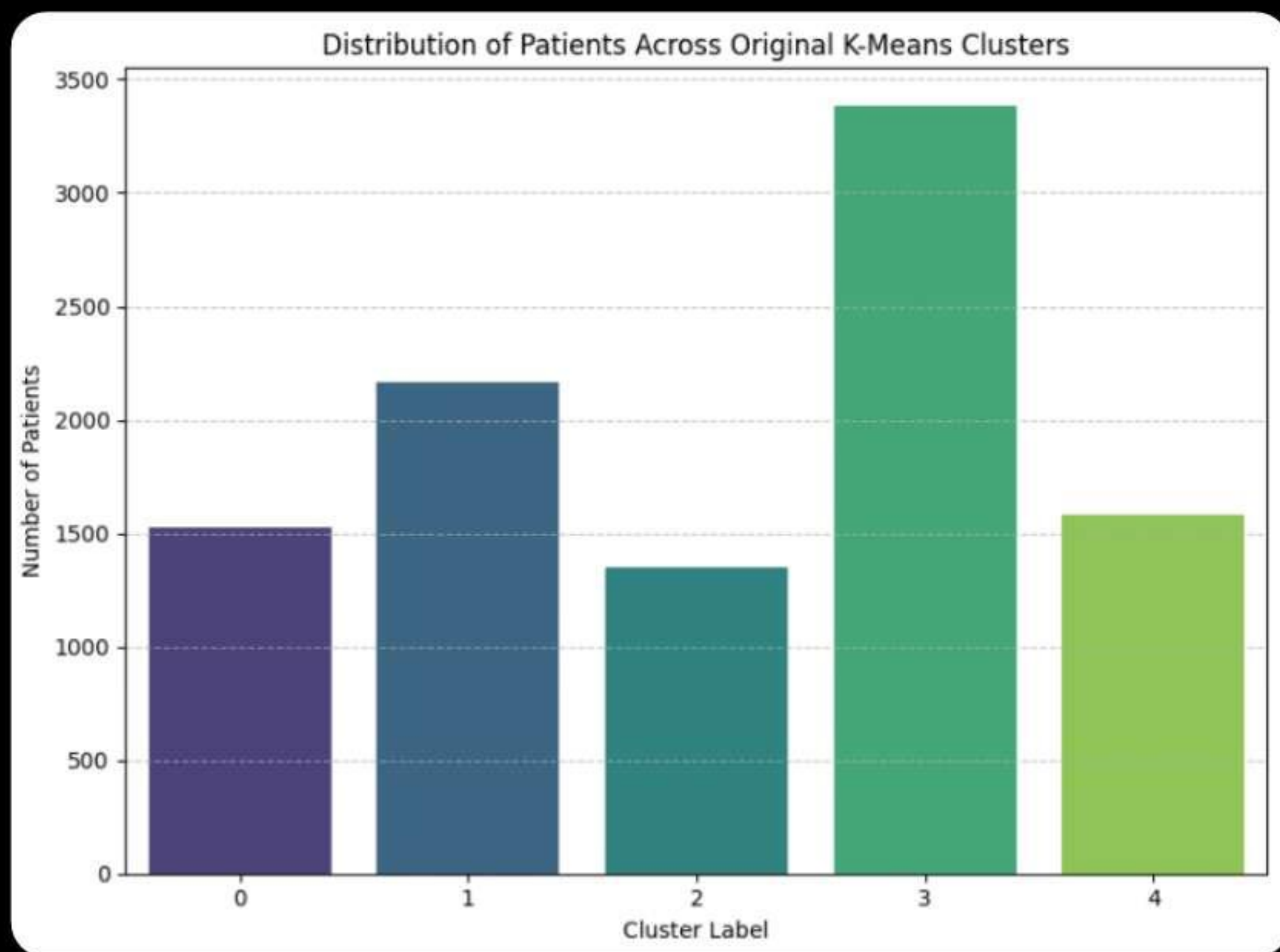
- Comorbidity_SDInteraction
- Combining race, insurance, and readmission



how do we **accurately + ethically**
give patients a risk score?



K-means Clustering



Cluster 1: High Risk: Highest comorbidities (4.08) and ED visits (3.34), Mostly emergency admissions (0.49)

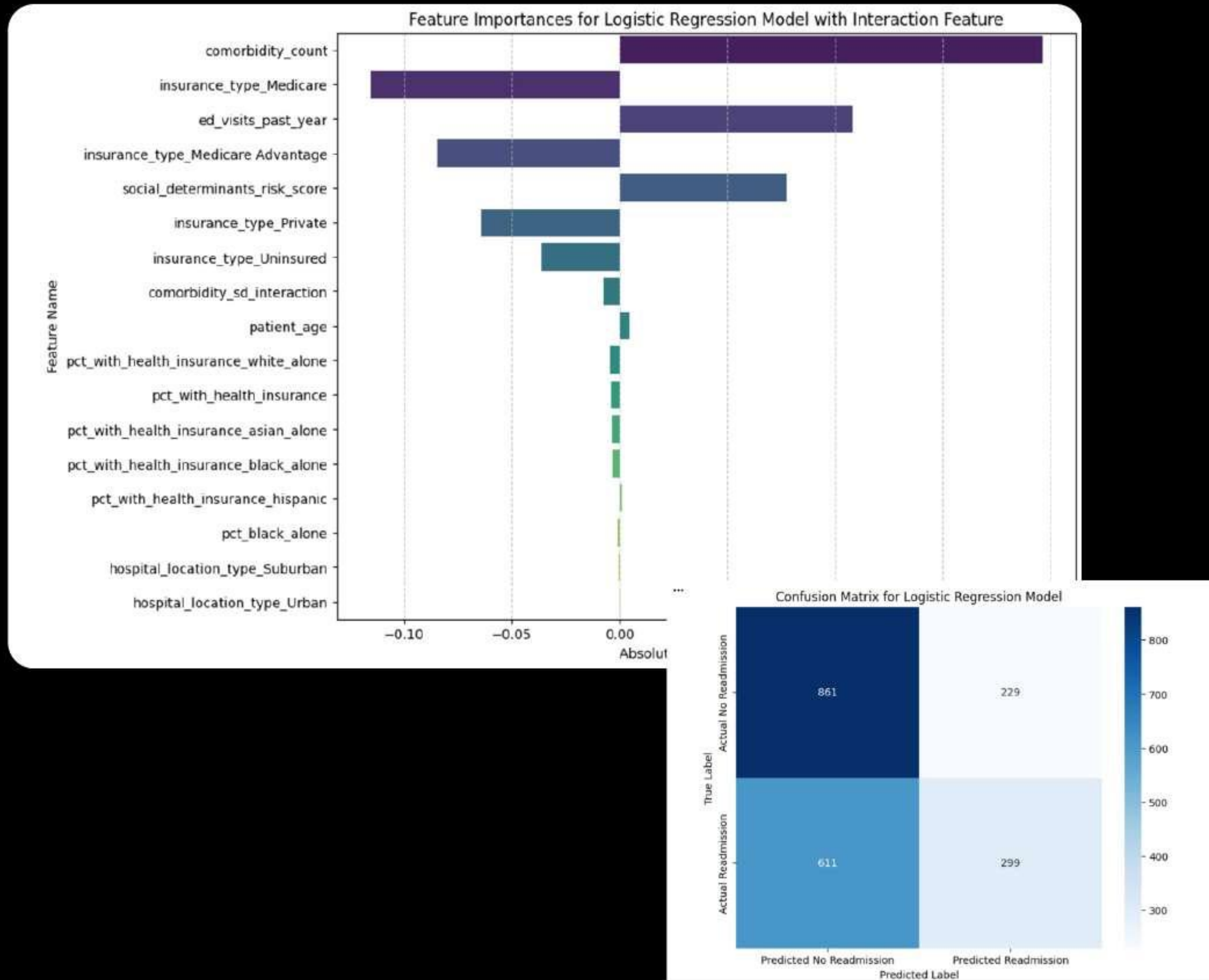
Cluster 2: Low Risk: Lowest comorbidities (0.60) and ED visits (0.58)

Cluster 3: Medium Risk (no follow up): Low comorbidities (1.16) and ED visits (0.62), No follow-up (0.0) is the primary risk factor despite low clinical burden

Cluster 4: Medium Risk: Moderate comorbidities (2.72) with very low ED visits (0.50), High follow-up (0.85) suggests controlled chronic disease (CV 0.27)

Cluster 5: Medium-High Risk: Moderate comorbidities (1.74) but high ED visits (2.40), High follow-up (0.78) yet frequent acute care signals elevated risk

Logistic Regression



Predicting Readmission 30 day based on comorbidity count, insurance type, and more

- Comorbidity count is the highest indicator of readmission rate, followed by having Medicare insurance and ED visits in the past year
- Rural patients also have the highest readmission rate

Confusion Matrix (indicates high performance)

- 861 True Negatives: The model correctly predicted 861 patients would not be readmitted, and they weren't.
- 299 True Positives: The model correctly predicted 299 patients would be readmitted, and they were.

Findings

Solution

01.

Most patients needing readmission are in the Third Cluster (Medium Risk with no-follow up)

02.

Comorbidity Count is the most important factor contributing to readmission rate

03.

Race does not have a major impact, but region and access to insurance does

Focus on Preventative Care for patients living in rural areas who's cluster category is in medium or high risk and may have a comorbidity.

Preventative Care for Cluster 3

- Individuals who are classified as medium risk (no follow up) in rural areas will be especially recommended to seek preventative care
- Implement this through targeted reminders in patient portal and through direct interaction with doctor
- These patients are often overlooked for not having the most severe or chronic conditions
- Funnel them into a partner specialist/PCP

Our Solution

Implementation Roadmap

March 2026 - April 2026

Classify current patients into clusters based using the k-means clustering model

Identify rural areas biweekly where > 50% of patients are in cluster 3 or 1 to start preventative care

May 2026 - August 2026

Instruct hospitals to send a reminder to seek preventative care through PCP or specialist

Teach physicians to communicate the new policy to those who are classified in cluster 3 or 1

Sept 2026 - Future

Reevaluate patient's categorized cluster every week through automated ML analysis

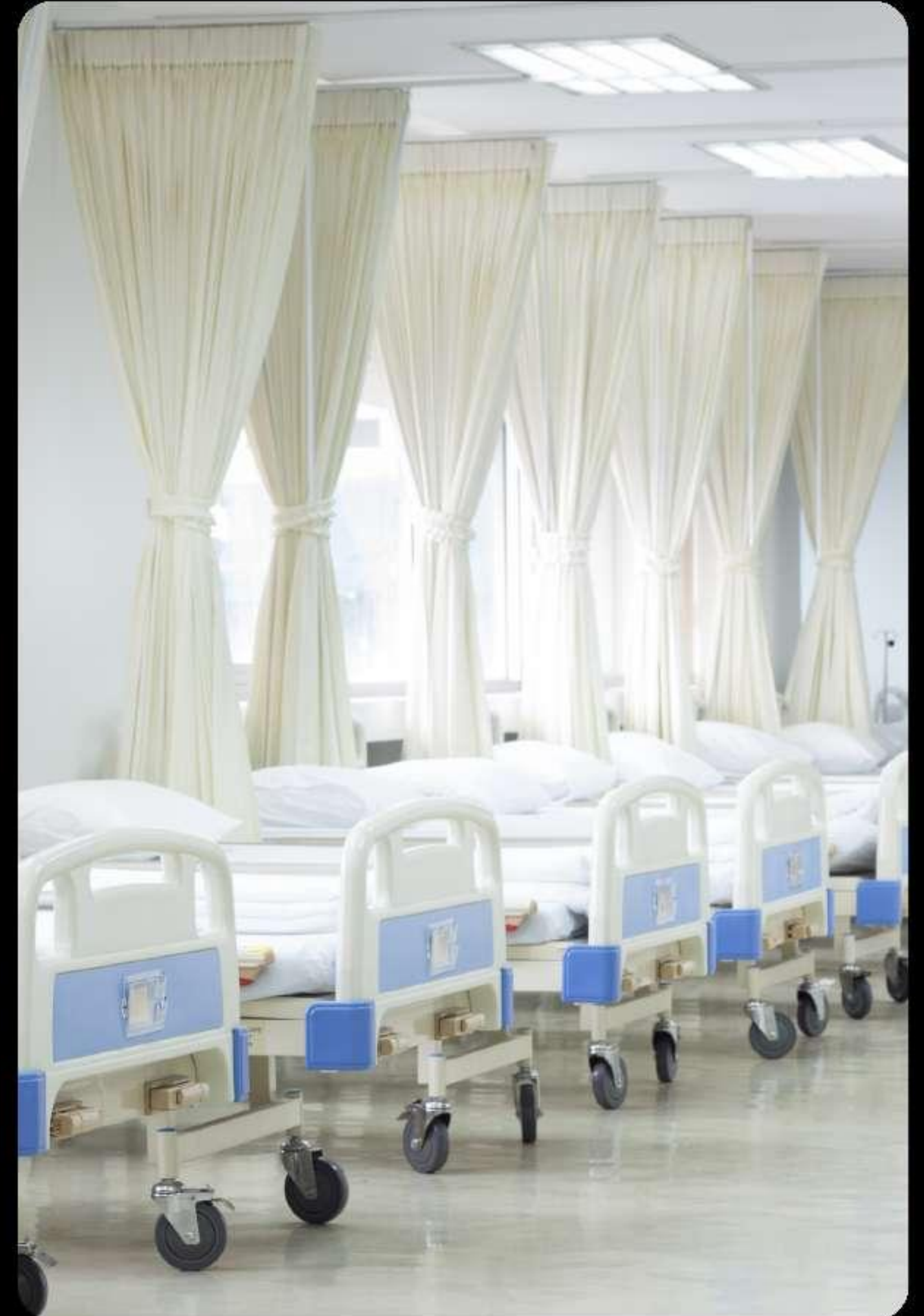
Compare results of new method to old approach and consider updating clusters if need be



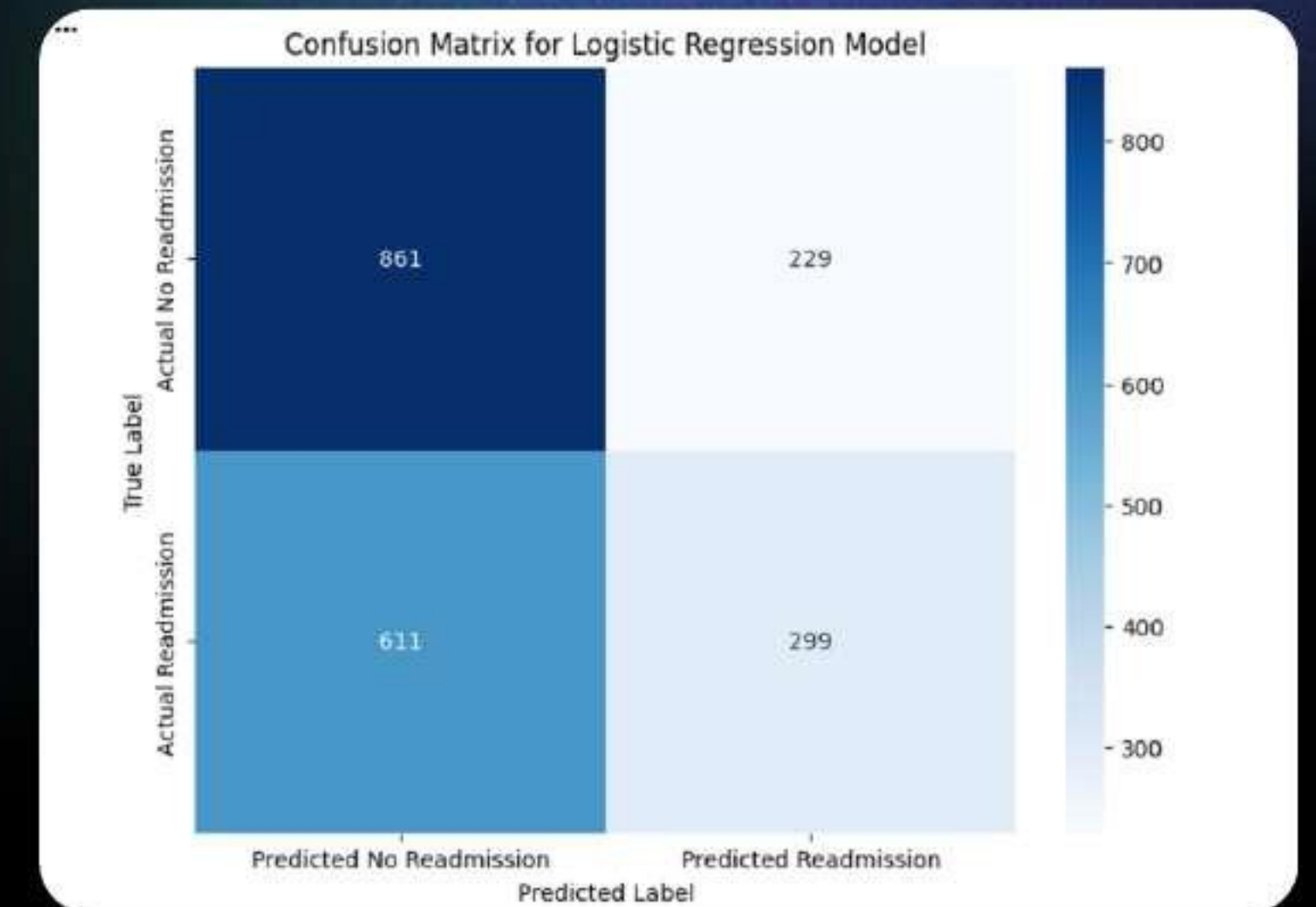
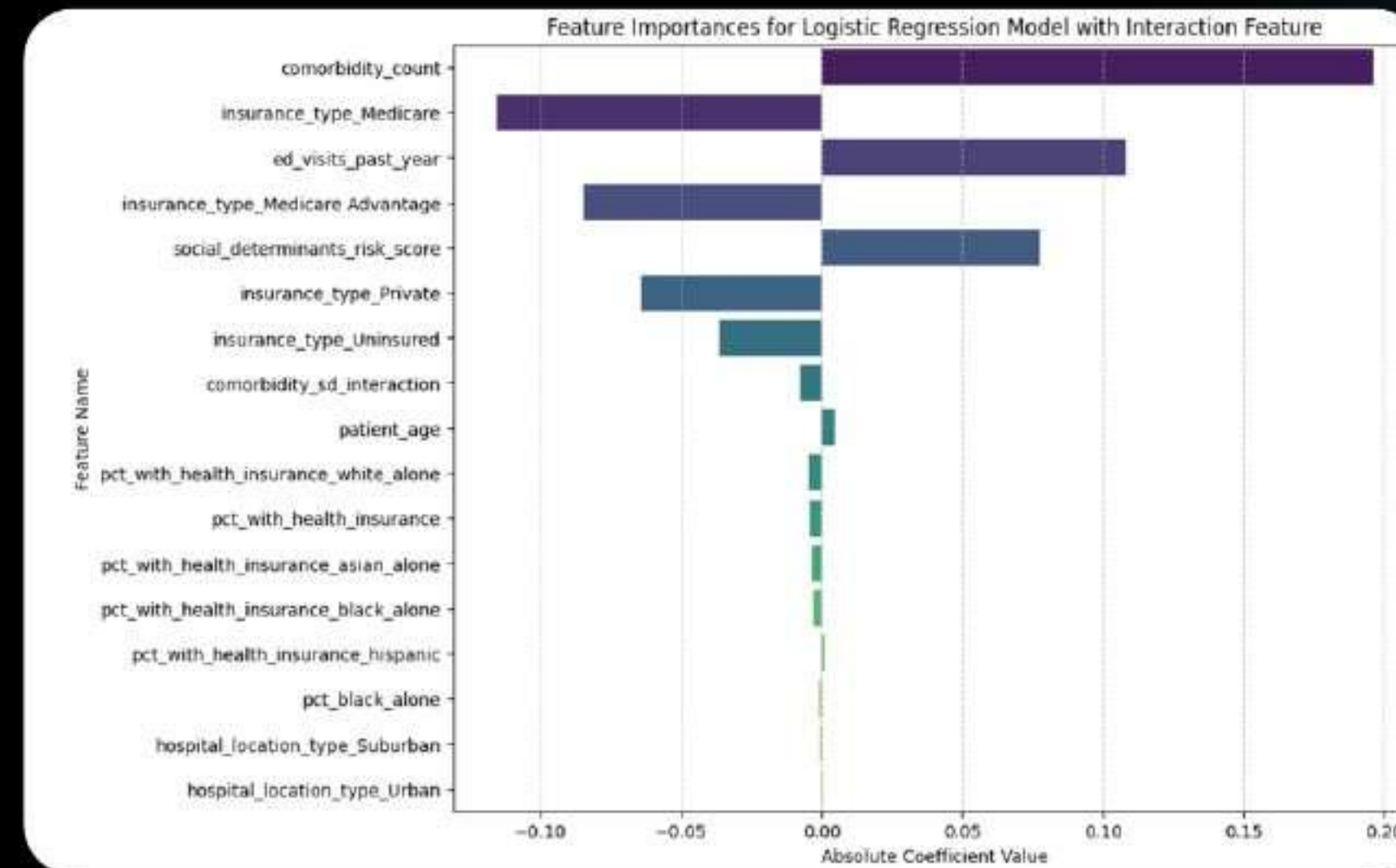
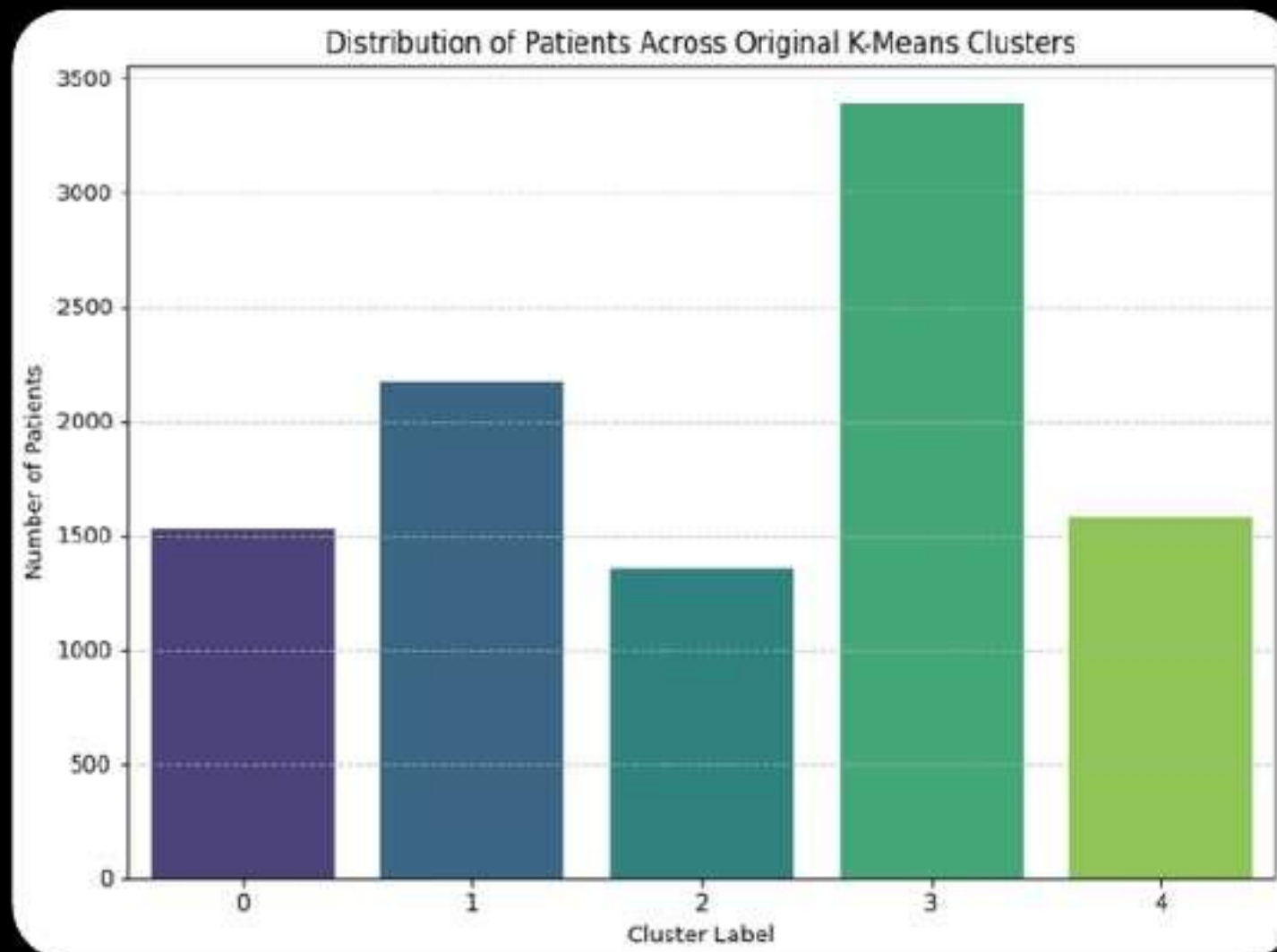
Conclusion

Through our detailed data analysis, we have effectively identified a major contributor to readmission rates.

We're tackling this through our patient classification system, giving hospitals a chance to spend time prescribing preventative care to those who need it the most + reduce readmission rates

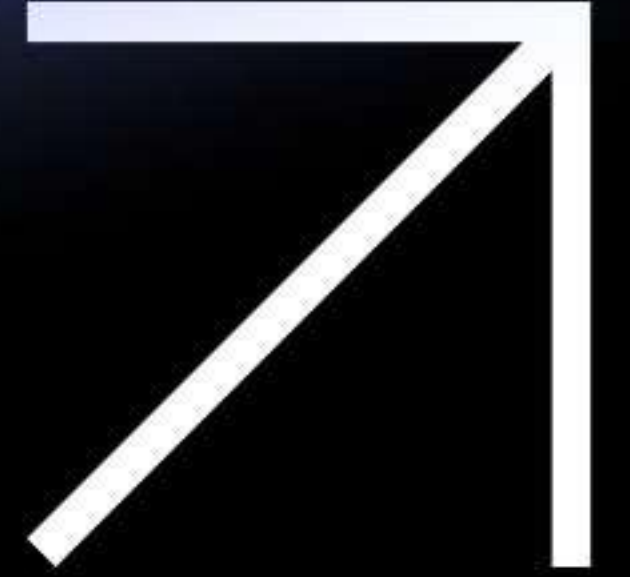


Appendix



all graphs built using Pandas in Google Colab with the help of Gemini

References



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Code for ML Models was done in part with the help of Gemini AI in Google Colab